AI4EAC Education Practice Challenge

1500 Zindi Points

Helping Kenya, Burundi and 6 other countries

Closing soon! (5 days left)

Skills you will learn Prediction Classification Feature Engineering

53 joined 20 active

Start	Close	Reveal
06 Mar 26	27 Mar 26	29 Mar

Can you identify which features of an early learning programme predict better learning outcomes for children?

DataDrive2030 Early Learning Predictors Challenge Introduction by Sonja Giese

Info Data Chat Leaderboard Team Submissions Submit

The Thrive by Five Index (2021) found that less than half of children attending an early learning programme (such as a preschool or creche) in South Africa start school with the right early learning foundation. There are many factors that influence whether a child Thrives by Five, such as access to a quality early learning programme, as well as poverty, gender, malnutrition, and emotional well-being. Children without a good foundation struggle to keep up at school and have a major learning disadvantage.

The Index used the Early Learning Outcomes Measure (ELOM) to assess children, and categorises their development as either “on track”, “falling behind” or “falling far behind”.

In this competition, your challenge is to use machine learning techniques to identify which early learning programme factors contribute to better learning outcomes in children, by predicting a child’s ELOM score.

This will allow DataDrive2030 to design better interventions that make optimal use of limited resources to ensure South Africa’s children are thriving.

About DataDrive2030 (datadrive2030.co.za)

Established in March 2022, DataDrive2030 is a South African based social enterprise that supports the collection and use of high quality data to drive improved child outcomes in the first 6 years of life. Our suite of early learning measurement tools accurately measure a range of developmental outcomes in young children, and provide an indication of the quality of the early learning environment in home and programme settings. Tools are digitised and designed for affordable use at scale in all official South Africa languages, with built-in data quality assurance mechanisms.

We aim to make these tools widely accessible, and to ensure that the information that is generated is easily understandable and mo importantly, is **actionable.** Using data, our goal is to drive tangible improvements in early childhood development services in South Africa, by 2030.

Evaluation

The error metric for this competition is the Root Mean Squared Error.

For every row in the dataset, submission files should contain 17 columns: ID, ELM Score and the 15 top predictors.

Your submission file should look like this:

```
child_id       target   feature_1             feature_2   ...
ID_AWX6YB4QO   56.999   child_score_item_15   child_score_item_7   ...
ID_SPYK410VL   64.070   child_score_item_4    child_score_item_3   ...
```

Prizes

There are 1 500 Zindi points available. You can read more about Zindi points here.

Rules

This challenge is open to anyone in:

- Burundi
- Democratic Republic of Congo
- Kenya
- Rwanda
- Somalia
- South Sudan
- Tanzania

- Uganda

Teams and collaboration

You may participate in competitions as an individual or in a team of up to four people. When creating a team, the team must have a total submission count less than or equal to the maximum allowable submissions as of the formation date. A team will be allowed t maximum number of submissions for the competition, minus the total number of submissions among team members at team formation. Prizes are transferred only to the individual players or to the team leader.

Multiple accounts per user are not permitted, and neither is collaboration or membership across multiple teams. Individuals and th submissions originating from multiple accounts will be immediately disqualified from the platform.

Code must not be shared privately outside of a team. Any code that is shared, must be made available to all competition participant through the platform. (i.e. on the discussion boards).

The Zindi data scientist who sets up a team is the default Team Leader but they can transfer leadership to another data scientist on the team. The Team Leader can invite other data scientists to their team. Invited data scientists can accept or reject invitations. Unt a second data scientist accepts an invitation to join a team, the data scientist who initiated a team remains an individual on the leaderboard. No additional members may be added to teams within the final 5 days of the competition or last hour of a hackathon.

The team leader can initiate a merge with another team. Only the team leader of the second team can accept the invite. The default team leader is the leader from the team who initiated the invite. Teams can only merge if the total number of members is less than equal to the maximum team size of the competition.

A team can be disbanded if it has not yet made a submission. Once a submission is made individual members cannot leave the tear

All members in the team receive points associated with their ranking in the competition and there is no split or division of the points between team members.

Datasets and packages

The solution must use publicly-available, open-source packages only.

You may use only the datasets provided for this competition. Automated machine learning tools such as automl are not permitted.

You may use pretrained models as long as they are openly available to everyone.

The data used in this competition is the sole property of Zindi and the competition host. You may not transmit, duplicate, publish, redistribute or otherwise provide or make available any competition data to any party not participating in the Competition (this includes uploading the data to any public site such as Kaggle or GitHub). You may upload, store and work with the data on any cloud platform such as Google Colab, AWS or similar, as long as 1) the data remains private and 2) doing so does not contravene Zindi's rules of use.

You must notify Zindi immediately upon learning of any unauthorised transmission of or unauthorised access to the competition dat and work with Zindi to rectify any unauthorised transmission or access.

Your solution must not infringe the rights of any third party and you must be legally entitled to assign ownership of all rights of copyright in and to the winning solution code to Zindi.

Submissions and winning

You may make a maximum of 10 submissions per day.

You may make a maximum of 300 submissions for this competition.

Before the end of the competition you need to choose 2 submissions to be judged on for the private leaderboard. If you do not mak selection your 2 best public leaderboard submissions will be used to score on the private leaderboard.

During the competition, your best public score will be displayed regardless of the submissions you have selected. When the competition closes your best private score out of the 2 selected submissions will be displayed.

Zindi maintains a **public leaderboard** and a **private leaderboard** for each competition. The Public Leaderboard includes approximately 20% of the test dataset. While the competition is open, the Public Leaderboard will rank the submitted solutions by the accuracy score they achieve. Upon close of the competition, the Private Leaderboard, which covers the other 80% of the test dataset, will be made public and will constitute the final ranking for the competition.

Note that to count, your submission must first pass processing. If your submission fails during the processing step, it will not be counted and not receive a score; nor will it count against your daily submission limit. If you encounter problems with your submissi file, your best course of action is to ask for advice on the Competition's discussion forum.

If you are in the top 10 at the time the leaderboard closes, we will email you to request your code. On receipt of email, you will have 48 hours to respond and submit your code following the **Reproducibility of submitted code** guidelines detailed below. Failure to respond will result in disqualification.

If your solution places **1st, 2nd, or 3rd on the final leaderboard**, you will be required to submit your winning solution code to us fo verification, and you thereby agree to assign all worldwide rights of copyright in and to such winning solution to Zindi.

If two solutions earn identical scores on the leaderboard, the tiebreaker will be the date and time in which the submission was mad (the earlier solution will win).

The winners will be paid via bank transfer, PayPal, or other international money transfer platform. International transfer fees will be deducted from the total prize amount, unless the prize money is under $500, in which case the international transfer fees will be covered by Zindi. In all cases, the winners are responsible for any other fees applied by their own bank or other institution for receiving the prize money. All taxes imposed on prizes are the sole responsibility of the winners. The top 3 winners or team leaders will be required to present Zindi with proof of identification, proof of residence and a letter from your bank confirming your banking details.Winners will be paid in USD or the currency of the competition. If your account cannot receive US Dollars or the currency of the competition then your bank will need to provide proof of this and Zindi will try to accommodate this.

Payment will be made after code review and sealing the leaderboard.

You acknowledge and agree that Zindi may, without any obligation to do so, remove or disqualify an individual, team, or account if Zindi believes that such individual, team, or account is in violation of these rules. Entry into this competition constitutes your acceptance of these official competition rules.

Zindi is committed to providing solutions of value to our clients and partners. To this end, we reserve the right to disqualify your submission on the grounds of usability or value. This includes but is not limited to the use of data leaks or any other practices that v deem to compromise the inherent value of your solution.

Zindi also reserves the right to disqualify you and/or your submissions from any competition if we believe that you violated the rule or violated the spirit of the competition or the platform in any other way. The disqualifications are irrespective of your position on th leaderboard and completely at the discretion of Zindi.

Please refer to the FAQs and Terms of Use for additional rules that may apply to this competition. We reserve the right to update these rules at any time.

Reproducibility of submitted code

- If your submitted code does not reproduce your score on the leaderboard, we reserve the right to adjust your rank to the score generated by the code you submitted.
- If your code does not run you will be dropped from the top 10. Please make sure your code runs before submitting your solutic
- Always set the seed. Rerunning your model should always place you at the same position on the leaderboard. When running yc solution, if randomness shifts you down the leaderboard we reserve the right to adjust your rank to the closest score that your submission reproduces.
- Custom packages in your submission notebook will not be accepted.
- You may only use tools available to everyone i.e. no paid services or free trials that require a credit card.

Documentation

A README markdown file is required

It should cover:

- How to set up folders and where each file is saved
- Order in which to run code
- Explanations of features used
- Environment for the code to be run (conda environment.yml file or an environment.txt file)
- Hardware needed (e.g. Google Colab or the specifications of your local machine)
- Expected run time for each notebook. This will be useful to the review team for time and resource allocation.

Your code needs to run properly, code reviewers do not have time to debug code. If code does not run easily you will be bumped down the leaderboard.

Consequences of breaking any rules of the competition or submission guidelines:

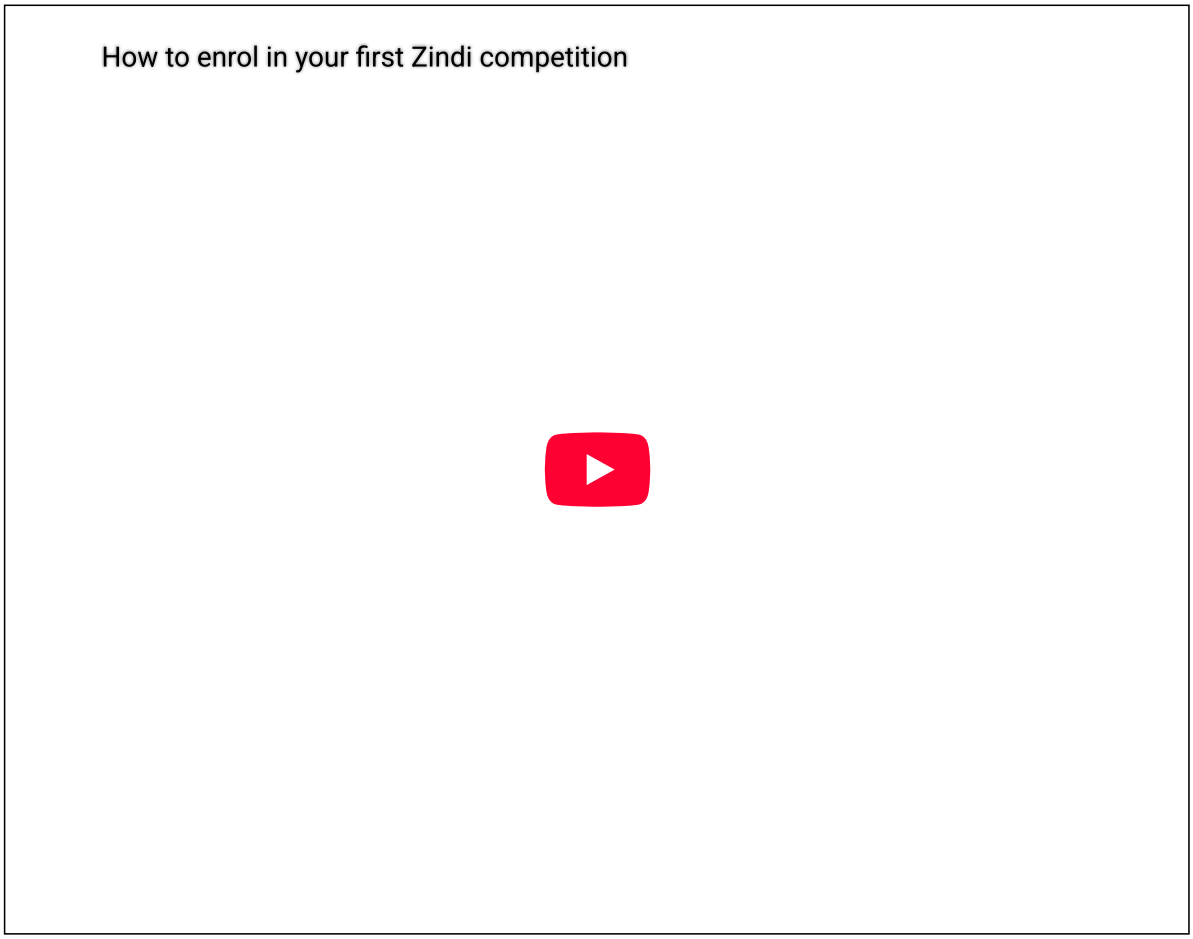
- **First offence: No prizes for 6 months and 2000 points will be removed from your profile (probation period).** If you are caught cheating, all individuals involved in cheating will be disqualified from the challenge(s) you were caught in and you will b disqualified from winning any competitions for the next six months and 2000 points will be removed from your profile. If you have less than 2000 points to your profile your points will be set to 0.
- **Second offence: Banned from the platform.** If you are caught for a second time your Zindi account will be disabled and you w be disqualified from winning any competitions or Zindi points using any other account.
- Teams with individuals who are caught cheating will not be eligible to win prizes or points in the competition in which the cheating occurred, regardless of the individuals' knowledge of or participation in the offence.
- Teams with individuals who have previously committed an offence will not be eligible for any prizes for any competitions durin the 6-month probation period.

Monitoring of submissions

- We will review the top 10 solutions of every competition when the competition ends.
- We reserve the right to request code from any user at any time during a challenge. You will have 24 hours to submit your code following the rules for code review (see above). Zindi reserves the right not to explain our reasons for requesting code. If you d not submit your code within 24 hours you will be disqualified from winning any competitions or Zindi points for the next six months. If you fall under suspicion again and your code is requested and you fail to submit your code within 24 hours, your Zin account will be disabled and you will be disqualified from winning any competitions or Zindi points with any other account.

How to get started with Zindi

How to enroll in your first Zindi competition



How to create a team on Zindi

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How to update your profile on Zindi

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ZINDI

Join the largest network for data scientists and AI builders

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